Challenge: Light Seeker

Points: 150 • Estimated Time: 90 Minutes

Challenge Scope

Using light sensors, the robot must locate a flashlight source placed at the far end of a darkened room.

Instructions & Engineering Hints

• **Objective:** Program your robot to autonomously navigate a darkened room, locate a stationary flashlight at the far end, and stop as close to the light source as possible without crashing.

• **Sensor Placement & Architecture:** Think carefully about where you mount your light sensors. If you use a single sensor, how will the robot know whether the light is to the left or the right? Consider using two light sensors placed symmetrically on the front of your chassis to compare light intensity readings between the left and right sides.

• **Algorithmic Approach:** Instead of hardcoding movements, design a continuous loop where the robot reads the light values, calculates the difference between the left and right sensors, and adjusts its steering dynamically. If the right sensor detects more light, the robot should steer right; if the left sensor detects more, it should steer left.

• **Thresholds & Calibration:** Ambient light in the room can fluctuate. Start your program by taking a baseline ambient light reading, or set a clear threshold value that tells the robot it has successfully reached the destination when the light intensity exceeds a specific limit.